

Asymptotic expansions of Maxwell-Jüttner moments and Haug 1988 and integrals: Detailed derivation.

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ABSTRACT

We present a detailed derivation of the asymptotic expansions and of the numerical algorithms adopted to evaluate the integrals appearing in the evaluation of the cross section σ_e and energy loss $d\varepsilon/dt$, appearing in Haug (1988), eqs. 27 and 15, respectively.

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1 ASYMPTOTIC EXPANSIONS FOR CROSS SECTION AND ENERGY LOSS

The integrals appearing in the expressions for the energy loss rate and of the cross section (Haug, 1988, eq. 27 and eq. 15, respectively) can be written in the following form:

$$\int_1^\infty d\gamma_r [g \text{ or } q] \left\{ \exp \left[-\frac{1}{\tau} h_- (\gamma_r, \varepsilon, p) \right] - \exp \left[-\frac{1}{\tau} h_+ (\gamma_r, \varepsilon,) \right] \right\} \quad (1)$$

where the functions g , q and $h_\pm (\gamma_r, \varepsilon, p, \gamma_r)$ are defined as:

$$g(x) = \frac{x^2}{\sqrt{x^2 - 1}(x - 1)} \quad (2)$$

$$q(x, \varepsilon, p | \tau) = g(x) \cdot \left[\varepsilon(x - 1) - p\sqrt{x^2 - 1} + \tau \right] \cdot F(\sin \theta_m^*, x) \quad (3)$$

$$F(\sin \theta_m^*, x) = 1 - \ln(2 \sin^2 \theta_m^*) + \left(\frac{x-1}{x} \right)^2 \left(\ln 2 + \frac{1}{8} \right) \quad (4)$$

and:

$$h_\pm(x, \varepsilon, p) = \left(\varepsilon x \mp p\sqrt{x^2 - 1} \right) \quad (5)$$

We see that for fixed ε (or equivalently p) the function $h_+(\gamma_r)$ is monotonically increasing, thus in the limit $\tau \ll 1$ we will have $\exp\{-h_+/\tau\} \ll 1$ and we can neglect its contribution to the integral.

We will thus search asymptotic expansions of the integrals:

$$\int_1^\infty d\gamma_r [g \text{ or } q] \exp \left\{ -\frac{h_-(\varepsilon, p, \gamma_r)}{\tau} \right\} \quad (6)$$

in the limit $\tau \ll 1$. This has the form of a *Laplace integral*:

$$H_{g \text{ or } q}(\lambda) = \int_a^b dt [g \text{ or } q] e^{-\lambda \phi(t)} \quad (7)$$

with the positions $\lambda \equiv 1/\tau$, $\phi \equiv h_-$.

We will present a derivation of the asymptotic expansion of this integral around the *stationary point* where $\phi'(t_0) = 0$ up to the second order in the Taylor expansion, one order beyond the expansion presented in Bleistein and Handelsman (1975, chap. 5). This presentation will be a little bit more pedagogical than

usual.

First, we assume that $g(t)$, $q(t)$ and $\phi(t)$ admit continuous derivatives up to the fourth in the semi infinite interval we are interested in: $t \in [1, +\infty[$ and that the function $\phi(t)$ has a true minimum at some point $t_0 \in (1, +\infty[$ where $\phi_0'' \geq 0$. Following Bleistein and Handelsman (1975) we assume $\phi(t)$ to be monotonic in the semi intervals $[1, t_0]$ and $(t_0, \infty[$ and we introduce the new variable:

$$\rho = \phi(t) - \phi_0$$

Note that: $\rho \geq 0$ for $t \in [1, \infty[$. The integral 7 can now be rewritten as: $H(\lambda) = I_a(\lambda) + I_b(\lambda)$ where:

$$I_a(\lambda) = -\exp(-\lambda\phi_0) \int_0^{\phi(a)-\phi_0} d\rho L(\rho) \exp(-\lambda\rho) \quad (8)$$

$$I_b(\lambda) = \exp(-\lambda\phi_0) \int_0^{\phi(b)-\phi_0} d\rho L(\rho) \exp(-\lambda\rho) \quad (9)$$

where we have defined:

$$L(\rho) = \frac{g(t)}{\phi'(t)} \Big|_{t=\phi^{-1}(\phi_0+\rho)} \quad (10)$$

We will first consider the Taylor expansion of $L(\rho)$ around ϕ_0 in eq. 10, starting from the Taylor expansions of $g(t)$ and $\phi(t)$ up to the second order in the right semi-interval $(t_0, b]$:

$$g(t) = \sum_{i=0}^2 a_i (t-t_0)^i + O[(t-t_0)^2] \quad (11)$$

$$\phi(t) = \phi_0 + \sum_{k=0}^2 b_k (t-t_0)^{k+2} + O[(t-t_0)^4] \quad (12)$$

The integrand $L(t)$ can be expanded in a Taylor series¹:

$$L = \frac{1}{t-t_0} \frac{\sum_{i=0}^2 a_i (t-t_0)^i + O[(t-t_0)^2]}{\sum_{k=0}^2 (k+2) b_k (t-t_0)^k + O[(t-t_0)^2]} \quad (13)$$

The Taylor expansion of $L(t)$ up to the second order will then be obtained by dividing the Taylor series expansions above up to the second order in $(t-t_0)$, $L(t) = (t-t_0)^{-1} \sum_{l=0}^2 c_l (t-t_0)^l + O[(t-t_0)^2]$, and the resulting coefficients are:

$$\begin{aligned} c_0 &= \frac{a_0}{2b_0} \\ c_1 &= \frac{a_1(2b_0) - a_0(3b_1)}{4b_0^2} \\ c_2 &= \frac{a_2(4b_0^2) - a_1b_0(3b_1) + a_0b_0(9b_1^2) - c_0(4b_2)(4b_0^2)}{8b_0^3} \end{aligned}$$

which can be more conveniently written as:

$$c_0 = \frac{g_0}{\phi_0''} \quad (14)$$

$$c_1 = \frac{1}{\phi_0''} \left(g_0' - g_0 \frac{\phi_0'''}{2\phi_0''} \right) \quad (15)$$

$$c_2 = \frac{1}{2\phi_0''} \left[g_0'' - g_0 \frac{\phi_0'''}{3\phi_0''} - \left(g_0' - \frac{g_0}{2} \frac{\phi_0'''}{\phi_0''} \right) \frac{\phi_0'''}{\phi_0''} \right] \quad (16)$$

¹We implicitly assume that the argument is evaluated at $t = \phi^{-1}(\phi_0 + \rho)$

In a neighborhood of ϕ_0 we have: $\rho \equiv \phi(t) - \phi_0 = (1/2)\phi_0''(t - t_0)^2 + O(t - t_0)^2$, and inverting this we get:

$$t - t_0 = \begin{cases} \sqrt{2\rho/\phi_0''} & t \geq t_0 \\ -\sqrt{2\rho/\phi_0''} & t < t_0 \end{cases}$$

We will now use this expansion to express the function $L(\rho)$ in the integrand in eqs. 8 and 9 in terms of ρ . In the interval $[t_0, \infty[$ we get:

$$L(\rho) = \sqrt{\frac{\phi_0''}{2}} \frac{1}{\rho^{1/2}} \left[c_0 + c_1 \sqrt{\frac{2}{\phi_0''}} \rho^{1/2} + c_2 \left(\frac{2}{\phi_0''} \right) \rho \right] \quad (17)$$

Similarly the expansion in the interval $[a, t_0)$ turns out to be:

$$-L(\rho) = \sqrt{\frac{\phi_0''}{2}} \frac{1}{\rho^{1/2}} \left[c_0 - c_1 \sqrt{\frac{2}{\phi_0''}} \rho^{1/2} + c_2 \left(\frac{2}{\phi_0''} \right) \rho \right] \quad (18)$$

We now can plug these expression in eqs. 8, 9, computing the integrals over each semi interval of the three contributions from the two expression for $L(\rho)$. When computing the integrals we will consider the asymptotic limit $\lambda \rightarrow \infty$, or equivalently that the upper extreme of the integration interval extends to infinity. A detailed derivation is presented in Bleistein and Handelsman (1975, sec. 5.1), where one shows that the combination of the odd terms in ρ (those multiplying c_1) cancel out, thus the final asymptotic expansion up to $\lambda^{-5/2}$, i.e. with a remainder of the order $O(\exp\{-\lambda\phi_0\}/\lambda^{5/2})$ is:

$$H(\lambda) = I_a(\lambda) + I_b(\lambda) = \exp\{-\lambda\phi_0\} \sqrt{\frac{\phi_0''}{2}} 2\pi^{1/2} \left[\frac{1}{\lambda^{1/2}} c_0 + \frac{1}{\lambda^{3/2}} c_2 \right] \quad (19)$$

The odd terms in the integration variable multiplying the c_1 coefficient cancel out. When we substitute now the coefficients c_0, c_2 we found above (eqs. 16) we eventually get:

$$H(\lambda) = I_a(\lambda) + I_b(\lambda) = \exp\{-\lambda\phi_0\} \left[\left(\frac{2\pi}{\lambda\phi_0''} \right)^{1/2} g_0 + \left(\frac{\pi}{2\lambda^3\phi_0''} \right)^{1/2} \left\{ g_0'' - g_0 \frac{\phi_0'''}{3\phi_0''} - \left(g_0' - \frac{g_0}{2} \frac{\phi_0'''}{\phi_0''} \right) \frac{\phi_0'''}{\phi_0''} \right\} \right] \quad (20)$$

The first term in the r.h.s. of this equation coincides with eq. (5.1.21) of Bleistein and Handelsman (1975), while the second term represents the higher order contribution in $1/\lambda$. Eq. 20 is the asymptotic expansion of the more general integral eq 6.

1.1 Application to Haug's equations.

We will now apply the results of the preceding section to obtain asymptotic expansions for the cross section and the energy loss. The integral appearing in the cross section σ_e can be evaluated using eq 6 with the function g given by eq 2. The function h_- has a minimum for $\gamma_{r*} = \varepsilon h'_-(\gamma_{r*}) = 0$, and one easily verifies that:

$$h''_x = \frac{p}{\sqrt{x^2 - 1}} - \frac{px^2}{(x^2 - 1)^{3/2}}$$

and when evaluated at the minimum $\gamma_{r*} \equiv \varepsilon$ one obtains: $h''(\varepsilon) = -1/p^2 < 0$ (we have used $p^2 = \varepsilon^2 - 1$), thus showing that $\gamma_{r*} = \varepsilon$ is a true minimum and can be used to estimate the asymptotic expansion.

In the cross section integral the function $g(t)$ in the Laplace integral is given by eq. 2 above, and the phase $\phi(t) \equiv h_-$. After some simple algebra we find:

$$g''(x) = \frac{2x^3 + 7x^2 + 4x + 2}{(x - 1)(x^2 - 1)^{5/2}} \quad (21)$$

We are now in a position to use eq. 20 to evaluate the integral $H(\lambda)$; in our case: $g_0 = g(\varepsilon)$ $\phi_0'' = h_-''(\varepsilon)$ and similarly for g_0'' . Eventually we obtain:

$$H_g(\tau) = (2\pi)^{1/2} \tau^{1/2} \exp\left(-\frac{1}{\tau}\right) \left(\frac{2\pi\tau}{p^2}\right)^{1/2} \left[\frac{\varepsilon^2}{(\varepsilon-1)p} + \tau \cdot \frac{2\varepsilon^3 + 7\varepsilon^2 + 4\varepsilon + 2}{2(\varepsilon-1)p^2} \right] \quad (22)$$

The cross section σ_e is given by (Haug, 1988, eq. 27):

$$\sigma_e = \frac{4\pi r_e^2}{\varepsilon p K_2(1/\tau) \theta_m^{*2}} H_g(\tau) \quad (23)$$

In the limit $\tau \ll 1$ the Bessel function of second kind has the asymptotic expansion: $K_2(1/\tau) \rightarrow (\pi/2)^{1/2} \tau^{1/2} \exp(-1/\tau)$, and putting all these expressions together we eventually arrive at the following expression for σ_e :

$$\sigma_e = \frac{8\pi r_e^2}{(\varepsilon-1) \theta_m^{*2}} \frac{\varepsilon}{p} \left[1 + \tau \frac{2\varepsilon^3 + 7\varepsilon^2 + 4\varepsilon + 2}{2\varepsilon^2 p^2} \right] \quad (24)$$

For the *minimum scattering angle* θ_m^* we adopt the expression provided by Haug (1988), eq. 21a:

$$\theta_m^* = \frac{4\hbar}{m_e c D} \frac{\gamma_r}{(\gamma_r + 1) \sqrt{2(\gamma_r - 1)}}$$

The maximum impact parameter $D = \beta_{cm} c / v_{th}$, where $\beta_{cm} \approx \beta$ is the velocity of the electron-plasma center of mass and $v_{th} \approx (k_B T_c / \mu m_e)^{1/2} = c (\tau / \mu)^{1/2}$ is the thermal velocity, thus we eventually obtain: $D \approx \beta / \tau^{1/2} = p / (\varepsilon \tau^{1/2})$. Substituting in the expression above for θ_m^* we eventually arrive at:

$$\theta_m^* = \frac{4\hbar \tau^{1/2}}{\mu^{1/2} m_e c} \frac{\varepsilon^2}{p(\varepsilon + 1) \sqrt{2(\varepsilon - 1)}} \quad (25)$$

In all these expression the momentum $p \equiv p(\varepsilon) = \sqrt{\varepsilon^2 - 1}$. The dimensionless temperature $\tau = k_B T_c / m_e c^2 = 1.68637 \times 10^{-6} T_{c,4}$, where $T_{c,4} \equiv T / 10^4$ K.

In the case of the energy loss the derivation is similar, the main difference being that instead of the function $g(\gamma_r)$ given in eq. 2 we have inside the integral the expression for q from eq 3. The *phase* function is the same, $\phi(t) \equiv h_-(\gamma_r)$, and we know that it has a minimum at $\gamma_{r*} = \varepsilon$, where we have to evaluate the function q . Using the definition of the latter (eq. 3):

$$q(\gamma_{r*} = \varepsilon) = g(\varepsilon) \cdot \left[\varepsilon(\varepsilon - 1) - p\sqrt{\varepsilon^2 - 1} + \tau \right] = g(\varepsilon) \cdot \left[\varepsilon(\varepsilon - 1) - p^2 + \tau \right]$$

The energy loss equation is given by Haug (1988, eq. 15):

$$\frac{d\varepsilon}{dt} = \frac{\pi r_e^2 n_e}{\varepsilon p K_2(1/\tau)} H_q(\tau) \quad (26)$$

The asymptotic expansion of H_q can be obtained using the same procedure adopted for H_g :

$$H_q = (2\pi\tau)^{1/2} \frac{\exp(-1/\tau) \varepsilon p^2}{(\varepsilon - 1) \sqrt{\varepsilon^2 - 1}} \cdot \left[\varepsilon(\varepsilon - 1) - p\sqrt{\varepsilon^2 - 1} + \tau \right] \cdot F(\sin \theta_m^*, \varepsilon) \quad (27)$$

Also in this case we adopt the asymptotic expansion $K_2(1/\tau) \rightarrow (\pi/2)^{1/2} \tau^{1/2} \exp(-1/\tau)$; after having substituted in the equation above and few further reductions we arrive at:

$$\frac{d\varepsilon}{dt} = 2\pi r_e^2 n_e c \cdot \frac{\varepsilon}{\sqrt{\varepsilon^2 - 1} (\varepsilon - 1)} [1 + \tau - \varepsilon] \quad (28)$$

1.2 Domain of validity

The asymptotic expansions eqs. 24 and 28 has an algebraic singularity for $\varepsilon = 1$, inherited by a similar singularity in the integrand of equation 1. Following Haug (1988, sec. 3) a lower limit on the center of mass Lorentz factor γ_r arises from the existence of a maximum impact factor. This translates in a lower limit on $\varepsilon \approx \gamma_r - 1$:

$$\varepsilon \geq w + w^2 \quad (29)$$

whwre: $w = 2.05 \times 10^{-13} n_{e,10} \theta_L^{*-2} T_{c,4}^{-1}$.

We adopt this as a lower limit in the plots of the paper.

2 NUMERICAL EVALUATION OF INTEGRALS

The numerical evaluation of integrals in eq. 1 is complicated by the presence of algebraic singularities at the lower end of the integration interval, $\gamma_r = 1$. To overcome these difficulties we follow a twofold approach: simplify the integrand and apply a very stable numerical integration method. First, we decompose the function $g(\gamma_r)$ as follows:

$$g(\gamma_r) \equiv \frac{\gamma_r^2}{\sqrt{\gamma_r^2 - 1}(\gamma_r - 1)} = \sqrt{\frac{\gamma_r + 1}{\gamma_r - 1}} + \frac{1}{\sqrt{\gamma_r^2 - 1}(\gamma_r - 1)} \quad (30)$$

The first term contains an integrable singularity at $\gamma_r = 1$, which can easily be dealt with. In order to cope with the second term, we introduce a new variable: $z = (\gamma_r - 1) / (\gamma_r + 1)$, so that: $\gamma_r = (1 + z) / (1 - z)$, and we immediately see that with this change of variable the original semi-infinite integration interval is mapped on a finite one: $1 \leq \gamma_r \leq +\infty \rightarrow 0 \leq z \leq 1$. It is not difficult to verify that with this change of variable the second term of the integral becomes simpler:

$$\frac{1}{\sqrt{\gamma_r^2 - 1}(\gamma_r - 1)} d\gamma_r \rightarrow \frac{1}{2z^{3/2}} dz$$

In the Python function (method) SIGMA_EFF we adopt this decomposition and use the MPMATH² package, method MPQUAD with the default tanh-sinh algorithm (Bailey and Borwein, 2015) to perform both integrals.

3 ASYMPTOTIC EXPANSION OF MOMENTS OF MAXWELL-JÜTTNER.

The manifestly covariant Maxwell-Jüttner distribution function in a reference frame where the jet is seen to be moving at speed $\mathbf{V}_J$ is given by:

$$f_{MJ}(\gamma, p_{\parallel}) d^3 \mathbf{p} = A \exp\left(-\frac{\gamma}{\tau}\right) \exp\left(-\frac{\beta_J \gamma}{\tau} p_{\parallel}\right) \cdot \gamma d\gamma dp_{\parallel} d\phi \quad (31)$$

Here we have assumed f_{MJ} to be axisymmetric and made a change of variables $(p_{\perp}, p_{\parallel}, \phi) \rightarrow (\gamma, p_{\parallel}, \phi)$, and introduced the bulk dimensionless velocity $\beta_J = V_J/c$ and Lorentz factor $\gamma_J = (1 - \beta_J^2)^{-1/2}$ and $\tau = (k_B T / m_e c^2) \simeq T / (5.93 \times 10^9 K)$. This DF is now anisotropic, and we will derive a series of asymptotic expansions in the parameter $\lambda = 1/\tau \gg 1$ of different quantities and moments of it. We begin with the normalization factor A .

3.0.1 Asymptotic expansion of A

. The normalization condition reads:

$$1 = A \cdot \int_{\Omega_p} d^3 \mathbf{p} f_{MJ}(\mathbf{p}) \quad (32)$$

With the change of variables shown above the elementary volume in phase space transforms as: $p_{\perp} dp_{\perp} dp_{\parallel} d\phi \rightarrow \gamma d\gamma dp_{\parallel} d\phi$, the integral can be expressed as:

$$2\pi \int_1^{+\infty} d\gamma \gamma \exp\left[-\frac{\gamma}{\tau}\right] \int_{p_{\parallel, \min}}^{p_{\parallel, \max}} dp_{\parallel} \exp\left[\frac{\beta_J \gamma}{\tau} p_{\parallel}\right]$$

²<https://github.com/mpmath/mpmath>

To determine the limits $p_{\parallel,min}, p_{\parallel,max}$ we consider the equality: $p_{\parallel}^2 = \gamma^2 - 1 - p_{\perp}^2$, from which we easily derive: $p_{\parallel,min} = -(\gamma^2 - 1)^{1/2} \leq p_{\parallel} \leq (\gamma^2 - 1)^{1/2} = p_{\parallel,max}$. The inner integral can be easily computed, and we thus get:

$$1 = A \cdot 2\pi \left(\frac{\tau}{\beta_J \gamma_J} \right) \{G_- - G_+\} \quad (33)$$

where we have defined:

$$G_{\pm} = \int_1^{+\infty} d\gamma \gamma \exp \left[-\frac{\gamma_J}{\tau} \left(\gamma \pm \beta_J (\gamma^2 - 1)^{1/2} \right) \right] \quad (34)$$

We evaluate these integrals through asymptotic expansions for $\tau \ll 1$. These integrals are very similar to those in the previous sections, and of the form:

$$I(\lambda) = \int_a^b dt f(t) \exp[\lambda \phi(t)]$$

In our case: $f(t) \equiv t$, $\lambda \equiv \gamma_J/\tau$ and $\phi(t) \equiv h_{\pm}(t) = -\left[t \pm \beta_J (t^2 - 1)^{1/2}\right]$, and the relevant interval: $\gamma \in [1, \infty)$.

The asymptotic expansions of G_+ , G_- for $\lambda \gg 1$ differ significantly, mostly because h_- has a maximum inside the integration interval, while h_+ attains its maximum at $t_* = 1$. However we can not apply Bleistein and Handelsman (1975, eq. 5.6.13) to deduce the asymptotic expansion for G_+ because the derivative of $h_+ = -\left[t + \beta_J (t^2 - 1)^{1/2}\right]$ diverges at $t = 1$. We get estimates about the size of G_+ by noting that the argument of the exponential is bound: $-\lambda (\gamma + \beta_J \gamma) \leq -\lambda \left[\gamma + \beta_J (\gamma^2 - 1)^{1/2}\right] \leq -\lambda \gamma$, which results into bounds on G_+ :

$$\frac{e^{-\lambda(1+\beta_J)}}{\lambda(1+\beta_J)} \leq G_+ \leq \frac{e^{-\lambda}}{\lambda}$$

For $\lambda \gg 1$ both bounds tend exponentially to zero : we can thus neglect G_+ in the following.

We will now derive an asymptotic expansion for G_- in the limit $\lambda \gg 1$, by using the Laplace method up to 2nd order in $1/\lambda$, (Bleistein and Handelsman, 1975, eq. 5.1.17):

$$G_- = I_-(\lambda) \simeq \left[\frac{2\pi}{|\phi^{(2)}|} \right]_{t=t_*}^{1/2} \frac{1}{\lambda^{1/2}} e^{\lambda \phi(t_*)} \left\{ f(t_*) + \frac{1}{\lambda} \left[-\frac{f^{(2)}}{2\phi^{(2)}} * \frac{f\phi^{(4)}}{8(\phi^{(2)})^2} + \frac{f^{(1)}\phi^{(3)}}{2(\phi^{(2)})^2} - \frac{5f(\phi^{(5)})^2}{24(\phi^{(2)})^3} \right]_{t=t_*} \right\} \quad (35)$$

where the upper index in parentheses denotes the order of derivation and t_* the point where the phase $\phi(t) \equiv -\left[t - \beta_J (t^2 - 1)^{1/2}\right]$ has a maximum: $\phi^{(1)}(t_*) = 0, \phi^{(2)}(t_*) < 0$. In our case we also have: $f(t) = t$, and thus: $f^{(1)} = 1, f^{(n)} = 0, n > 1$. The derivatives of $\phi(t)$ are given by:

$$\phi^{(1)} = -1 + \beta_J t / (t^2 - 1)^{1/2}, \phi^{(2)}(t) = 3\beta_J t / (t^2 - 1)^{5/2}, \phi^{(4)} = -3\beta_J (4t^2 + 1) / (t^2 - 1)^{7/2}$$

One easily verifies that: $t_* = \gamma_J = (1 - \beta_J^2)^{-1/2}$. A few further simple but tedious calculations give:

$$\phi(\gamma_J) = -\gamma_J^{-1}, \phi^{(2)}(\gamma_J) = -\beta_J^{-2} \gamma_J^{-3}, \phi^{(3)}(\gamma_J) = 3\beta_J^{-4} \gamma_J^{-4}, \phi^{(4)}(\gamma_J) = -3(4\gamma_J^2 + 1) / \beta_J^6 \gamma_J^7$$

After a few cumbersome algebraic steps we calculate the erm under square parentheses in eq 35 as: $15\gamma_J^2/8$. Note that in our case $\lambda = \gamma_J/\tau$, and thus:

$$G_- \sim (2\pi)^{1/2} \beta_J \gamma_J \tau^{1/2} \exp \left(\frac{1}{\tau} \right) \gamma_J \left\{ 1 + \frac{15}{8} \tau + O(\tau^2) \right\}$$

If we now insert this into eq. 33 we get the normalization constant:

$$A \sim \frac{\gamma_J^{-1} \left[1 + \frac{15}{8} \tau + O(\tau^2) \right]^{-1}}{(2\pi)^{3/2} \tau^{3/2} \exp(1/\tau)} \quad (36)$$

For $\beta_J \rightarrow 0$, $\gamma_J \rightarrow 1$ we see that: $A \rightarrow 1/(2\pi)^{3/2} \tau K_2(\tau)$ (neglecting the linear term in τ in parentheses), i.e. the value from the manifestly covariant formalism of Chacón-Acosta et al. (2010).

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